

Information Technology University(ITU).
Assignment-1: User Analysis & Interaction Problem Modeling

PandaSwift Grocery: An online grocery ordering system

Subject: Human computer interaction

Assignment 1

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1. Project overview (10 Marks)

Project title

PandaSwift Grocery: An online grocery ordering system: The Frictionless Grocery Experience

Domain: E-commerce / Hyper-local Retail Tech

Problem statement

Despite the growing popularity of online food delivery platforms grocery ordering systems often suffer from usability issues including:

- Complex navigation for large product catalogs
- Poor filtering and search efficiency
- out of stock frustration etc

This project aims to systematically analyze user needs and interaction challenges in order to design a high usability grocery e commerce system

Why this problem is important

- **Time sensitivity of grocery needs:** Groceries are essential goods. delays or poor UX directly impact daily life.
- **Diverse user base:** Users range from students and working professionals to elderly users and low-tech individuals. Poor interaction design excludes vulnerable groups.
- **Mobile first behavior:** Most users access systems through smartphones. Small screens increase navigation complexity.

Current limitations in existing systems

Even advanced platforms exhibit the following interaction problems:

- **Poor product discovery:** Inefficient filtering systems, search that fails with minor spelling errors, and weak categorization hierarchy.
- **Accessibility gaps:** Small clickable areas, low contrast UI elements, and lack of assistive support for elderly users.
- **Error recovery weakness:** Limited undo options, cart reset issues, and accidental order placement.

2. Target users and personas (20 Marks)

Persona 1: The Time Constrained Professional

- **Name:** Ayesha Khan
- **Age:** 27
- **Occupation:** Software engineer(Lahore)
- **Technical Proficiency:** High (advanced mobile app user)

Goals:

- Order weekly groceries within 5–10 minutes
- Quickly reorder frequently purchased items
- Track delivery accurately
- Avoid store visits due to busy schedule
- Use digital payments seamlessly

Pain points:

- Too many promotional banners causing distraction
- Slow search results
- Items shown available but later marked out-of-stock
- Hidden delivery charges during checkout
- Re entering address/payment repeatedly

Behavioral patterns:

- Uses app late at night (10 PM – 12 AM)
- Searches directly instead of browsing categories
- Frequently reorders previous items
- Prefers wallet payment
- Values time efficiency over price comparison

Accessibility considerations:

- Uses phone in low light conditions
- Often multitasking needs fast cognitive processing
- Requires clear confirmation messages to avoid accidental orders

Why this persona is relevant: This segment represents urban mobile first users who drive high transaction frequency. They expect high efficiency and low friction and are highly sensitive to UX delays. Design must optimize for fast search, smart recommendations, one-tap reordering, and minimal checkout friction.

Persona 2: The household budget manager

- **Name:** Imran Ahmed
- **Age:** 38
- **Occupation:** School Teacher
- **Technical Proficiency:** Moderate

Goals:

- Buy monthly groceries within a fixed budget
- Compare prices across brands
- Find discounts and bundle offers
- Ensure correct quantity selection
- Avoid unnecessary spending

Pain Points:

- Difficult price comparison across brands
- Confusing quantity units (kg vs grams)
- Accidental cart modifications

- Fear of payment errors
- Limited filtering options

Behavioral Patterns:

- Shops mostly on weekends
- Browses categories instead of searching
- Checks reviews before purchasing
- Compares multiple brands before adding to cart
- Prefers Cash on Delivery (trust factor)

Accessibility Considerations:

- Needs larger readable product information
- Requires clear price breakdown
- Needs confirmation before order placement
- May experience confusion with complex UI layouts

Why This persona is relevant: This segment represents middle-income family buyers who have larger cart sizes. They are price-sensitive, detail-oriented, and face high cognitive load during comparison. Design must optimize for a clear product hierarchy, transparent pricing, easy quantity modification, and budget visibility tools.

3. Context of use analysis (10 Marks)

1. Where will the system Be Used?

The system will primarily be used:

- At home (living rooms, kitchens, bedrooms)
- In offices during work breaks
- In hostels/apartments by students
- Occasionally while commuting

Since grocery shopping is a routine activity, usage is frequent and situational rather than planned.

2. On what device?

- **Primary:** Smartphones
- **Secondary:** Computers/laptops

first approach, considering:

- Small screen sizes
- Touch based interaction
- One handed usage

3. Environmental constraints

- Low lighting for night time

- Background noise TV, family conversations etc
- Time pressure like ordering urgently

These conditions increase:

- Cognitive load,Risk of accidental taps,Reduced attention span

4. Physical limitations

Potential user limitations include:

- One handed phone usage
- Visual strain
- Fatigue during long sessions .
- Elderly use

Therefore, UI must support:

- Large touch targets, Clear typography,Minimal scrolling effort

5. Social context

- Used individually for family decision making .
- Budget discussions
- Users may compare prices with family

6. Connectivity conditions

In Pakistan, connectivity may vary:

- 4G in urban areas while weak signals in some regions

The system must:

- Load quicklyby showing clear loading states
- Handle slow networks gracefully
- Prevent data loss during checkout

4. Task analysis (20 Marks)

System Context: PandaMart : Online Grocery Module

Task 1: Search for a Grocery Item	
Task Goal	Quickly find a specific product .
Steps	1.Open PandaMart 2. Tap search bar 3. Enter product name 4. View results 5. Select item
Possible Errors	Spelling mistakes or tapping wrong product.
System Response	Auto suggestions, spell correction, clear product thumbnails, highlight popular items.
Usability Risks	Poor search algorithm increases frustration.
Task 2: Browse & Compare Products	
Task Goal	Compare brands, prices, and quantities before purchasing.
Steps	1. Select category 2. Scroll product list 3. Open product details 4. Compare other options
Possible Errors	Confusion in units , misreading discount etc.
System Response	Clear price per unit, filter and sort options, visible proper discount labels.
Usability Risks	Long scrolling increases fatigue; poor categorization increases navigation confusion.
Task 3: Add Item to Cart & Modify Quantity	
Task Goal	Add desired items and adjust quantities accurately.
Steps	1. Tap “Add to Cart” 2. Increase/decrease quantity 3. View updated cart
Possible Errors	Adding wrong quantity, accidental multiple taps, forgetting item added.
System Response	Instant visual feedback, clear quantity indicator, undo option.
Usability Risks	Small touch targets gives accidental input and lack of feedback causes confusion.
Task 4: Checkout & Payment	
Task Goal	Complete order securely with preferred payment method.
Steps	1. Open cart 2. Review items 3. Confirm address 4. Select payment 5. Place order
Possible Errors	Wrong address selection, payment method confusion, accidental order placement.
System Response	Clear cost breakdown, confirmation screen, payment failure messages, edit options.
Usability Risks	Hidden charges reduce trust; complex checkout increases abandonment rate.
Task 5: Track Order & Handle Issues	
Task Goal	Monitor delivery status and resolve problems.
Steps	1. View order status 2. Track rider location
Possible Errors	Misinterpreting delivery status , difficulty finding help option.
System Response	Real time tracking, clear status labels and visible support button.
Usability Risks	Poor feedback increases anxiety; unclear support access harms user satisfaction.

5. Functional Requirements (10 Marks)

System: PandaSwift Grocery: Online Grocery Ordering Module

A. Core Features

- **User Authentication**
 - Allow users to sign up, log in, log out.
- **Product search and browsing**
 - Enable keyword based search.
 - Provide filtering and sorting
- **Product detail view**
 - Display product image, price, unit, description, availability etc .
 - Show discount and stock status.
- **Cart management**
 - Add/remove products.
 - Display real time cart total.
- **Order placement and Checkout**
 - Confirm delivery address.
 - Select payment method .
 - Display final price before doing confirmation
- **Order tracking**
 - Show order status
 - Provide delivery time.

B. Supporting features

- **Reorder previous purchases** Allow users to reorder items from order history.
- **Wishlist:** Save products for future purchase.
- **Promotions and discount display:** Show active deals and coupon options.
- **Order history:** View previous transactions and invoices.
- **Customer support access:** Provide chat/help center for issue resolution.

C. Interaction based functionalities

- **Auto suggestions in the search:** Provide real time search recommendations.
- **Real time feedback:** Show visual confirmation when an item is added to the cart.
- **Error handling:** Display clear error messages for payment failure or out of stock .

- **Dynamic cart updates:** Automatically update total price when quantity changes.
- **Loading and progress indicators:** Show system status during search, checkout and payment processing.

6. Non functional requirements (15 Marks)

1. Usability

Requirement:

- The system shall enable users to complete a grocery order within minimal steps
- Price breakdown must be visible before final confirmation.

Why important: grocery shopping involves large product sets and frequent use. High usability reduces cognitive load, prevents frustration and lowers cart abandonment rates.

2. Accessibility

Requirement:

- Provide readable font sizes and high color contrast and large buttons.
- Support screen readers and assistive technologies.

Why important: The system serves diverse users including elderly and moderate tech users. Accessibility ensures inclusivity and prevents interaction barriers.

3. Learnability

Requirement:

- First time users should understand core features
- Use familiar e commerce patterns

Why important: Users should quickly learn how to search, add items, and checkout.

4. Performance perception

Requirement:

- Search results should appear within 2 to3 seconds.
- Loading indicators must be shown during delays.

Why important: Fast feedback improves trust and reduces anxiety during payment or order tracking.

5. Error tolerance

Requirement:

- Provide undo options
- Confirm before final order placement.
- clear error messages

Why Important: Error tolerance prevents financial mistakes and increases user confidence.

6. Safety

Requirement:

- Secure payment processing and protect data
- Confirm high value transactions.

Why Important: Users must trust the platform when making payments.

7. Interaction challenges and risk analysis (10 Marks)

1. Cognitive load risks

- **Risk:** Large product catalogs, multiple brands and price comparisons, promotional banners and pop-ups, and complex checkout price breakdowns.
- **Impact on UX:** High cognitive load leads to decision fatigue, slower task completion, cart abandonment, and reduced satisfaction. If users feel overwhelmed, they may exit without purchasing.

2. Navigation confusion risks

- **Risk:** Poor category hierarchy, deep navigation levels, inconsistent back navigation, and hidden filters.
- **Impact on UX:** Users struggle to find products, leading to increased interaction time, frustration, and repeated searching. Navigation confusion directly reduces usability .

3. Error prone tasks

- **High risk tasks:** Selecting the wrong quantity, entering the wrong delivery address, accidental order placement, and payment errors.
- **Impact on UX:** Financial loss concerns, reduced trust, and a negative perception of reliability.

4. Accessibility barriers

- **Risk:** low contrast UI elements, small font sizes and button, and color-only discount indicators.
- **Impact on UX:** Exclusion of elderly users, higher input errors, increased visual strain . Accessibility gaps limit user reach and violate inclusive design principles.

5. Privacy and ethical concerns

- **Risk:** Storage of payment information, exposure of data, order history profiling and targeted promotional manipulation.
- **Impact on UX:** Reduced trust, fear of data misuse, hesitation in online payments

8. Application of HCI principles (5 Marks)

To address the above risks, the following HCI principles must guide the design:

1. Consistency

- Uniform button styles.
- Standard cart icon placement.
- Consistent checkout flow.

Benefit: Reduces learning time and prevents navigation confusion.

2. Feedback

- Visual confirmation when an item is added to the cart.
- Real-time order tracking updates.
- Clear payment success or failure messages.

Benefit: Reduces anxiety and increases system transparency.

3. Visibility

- Always visible cart summary.
- Clear stock availability indicators.
- Transparent price breakdown.

Benefit: Users remain aware of system status and costs at all times.

4. Affordance

- Buttons visually appear clickable.
- Quantity controls clearly indicate increase or decrease functions.
- Search bar is clearly identifiable.

Benefit: Reduces hesitation and interaction errors.

5. Error Prevention

- Confirmation before final order placement.
- Disable checkout if the cart is empty.
- Validate address and payment fields.

Benefit: Prevents costly mistakes and enhances user confidence.

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